Alternating Current

TARGET: NEET UG PHYSICS | NCERT CLASS 12 PHYSICS | COMPREHENSIVE THEORY & PRACTICE

1. Syllabus Overview & Learning Objectives

In direct current (DC) circuits, electric current flows steadily in one direction. However, electric energy supplied to homes, hospitals, and industrial centers is predominantly distributed as **alternating current (AC)**, where voltage and current vary sinusoidally with time. Alternating voltages can be stepped up or down efficiently using transformers, enabling economical power transmission over hundreds of kilometers with minimal Joule heating loss.

Key NEET UG Exam Focus Areas

- Mathematical descriptions of sinusoidal AC voltage, peak values, and root mean square (RMS) values.
- Phase relationships between alternating voltage and alternating current across pure resistors, inductors, and capacitors.
- Analytical and geometric phasor treatments of series LCR circuits, total impedance, and phase angle.
- The phenomenon of electrical resonance: resonant angular frequency, sharp current peaks, and dynamic characteristics.
- Average power dissipation, power factor ($\cos\phi$), wattless current, and power factor correction.
- Transformers: operational principles of mutual induction, turn ratios, step-up/step-down mechanisms, and sources of energy dissipation.

2. Alternating Quantities & Elementary AC Circuits

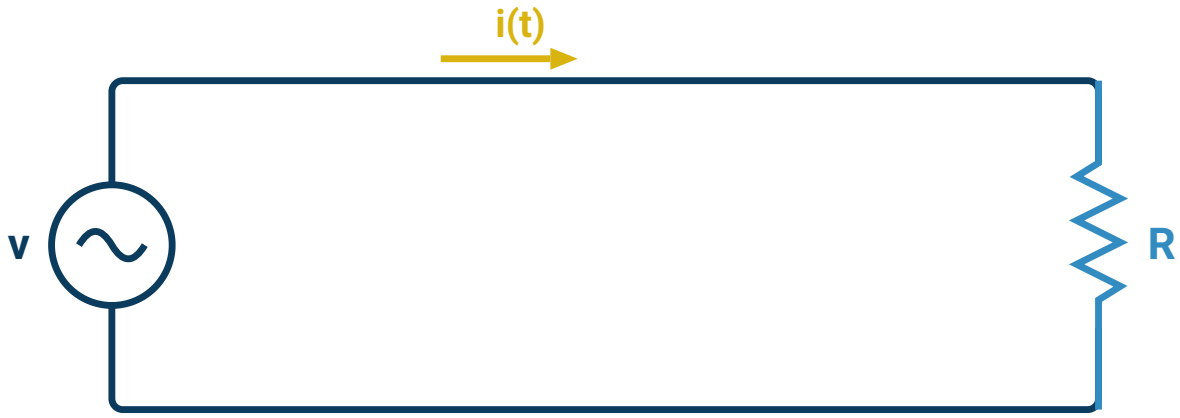
2.1 Sinusoidal Alternating Voltage and Current

An alternating voltage source produces a time-varying electric potential across its output terminals according to the harmonic relationship:

$$v = v_m \sin(\omega t)$$

where v_m is the peak voltage (voltage amplitude), $\omega = 2\pi\nu = \frac{2\pi}{T}$ is the angular frequency in $\mathrm{rad/s}$, ν is the frequency in Hertz (Hz), and T is the period of one complete cycle in seconds.

BOSONANCE



Tutorial Diagram 1: AC generator supplying a purely resistive circuit element.

2.2 AC Voltage Applied to a Pure Resistor

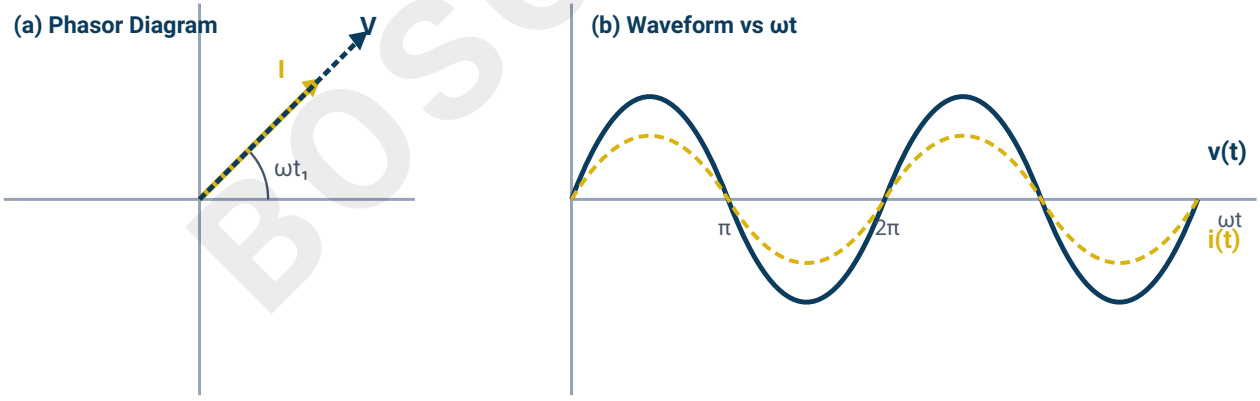
Applying Kirchhoff's loop rule around the circuit in Tutorial Diagram 1:

$$v_m \sin(\omega t) - iR = 0 \implies i = \frac{v_m}{R}\sin(\omega t) = i_m \sin(\omega t)$$

where the current amplitude is:

$$i_m = \frac{v_m}{R}$$

Comparing $v = v_m \sin(\omega t)$ and $i = i_m \sin(\omega t)$, we observe that across an ideal resistor, **the voltage and current oscillate in phase**. Both reach zero, minimum, and maximum values at identical moments in time.



Tutorial Diagram 2: In-phase relationship between voltage and current across a pure resistor.

2.3 Power Dissipation and Root Mean Square (RMS) Values

The algebraic sum of instantaneous current values over an entire cycle is identically zero:

$$\langle i \rangle = \frac{1}{T}\int_0^T i_m \sin(\omega t)\, \mathrm{d}t = 0$$

However, the dissipation of electrical energy as heat depends on Joule's heating law $i^2 R$. Because i^2 is always non-negative, electrical energy is continuously dissipated. The instantaneous power consumed in the resistor is:

$$p = i^2 R = i_m^2 R \sin^2(\omega t)$$

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Using the trigonometric identity $\sin^2(\omega t) = \frac{1}{2}[1 - \cos(2\omega t)]$, the average power $\bar{p}$ or P dissipated over one complete period T is:

$$P = \langle p \rangle = i_m^2 R \langle \sin^2(\omega t) \rangle = i_m^2 R \left(\frac{1}{2} - \frac{1}{2} \langle \cos(2\omega t) \rangle \right)$$

Since the time average of $\cos(2\omega t)$ over a cycle is zero:

$$\langle \cos(2\omega t) \rangle = \frac{1}{T} \int_0^T \cos(2\omega t)\, \mathrm{d}t = \frac{1}{2\omega T} [\sin(2\omega T) - 0] = 0$$

$$P = \frac{1}{2} i_m^2 R$$

To write this in the identical algebraic form as DC electrical power ($P = I^2 R$), we define the **root mean square (RMS)** or **effective current**:

$$I = I_{\text{rms}} = \sqrt{\langle i^2 \rangle} = \sqrt{\frac{i_m^2}{2}} = \frac{i_m}{\sqrt{2}} \approx 0.707\, i_m$$

Similarly, the RMS or effective voltage is defined as:

$$V = V_{\text{rms}} = \frac{v_m}{\sqrt{2}} \approx 0.707\, v_m$$

Consequently, Ohm's law and power dissipation in terms of RMS parameters take the elegant forms:

$$V = IR \quad \text{and} \quad P = I^2 R = \frac{V^2}{R} = VI$$

Physical Interpretation of the AC Ampere:
The AC ampere cannot be defined by the magnetic attraction between parallel conductors because the alternating force averages to zero over each cycle. Therefore, one ampere of RMS alternating current is defined as the current that produces the exact same average Joule heating in a given resistor under identical conditions as one ampere of steady direct current.

2.4 AC Voltage Applied to a Pure Inductor

Consider an ideal inductor with self-inductance L and negligible internal resistance connected to an alternating source $v = v_m \sin(\omega t)$. By Faraday's law of induction and Lenz's law, an opposing back emf $-L\frac{\mathrm{d}i}{\mathrm{d}t}$ is developed. Applying Kirchhoff's loop rule:

$$v - L\frac{\mathrm{d}i}{\mathrm{d}t} = 0 \implies \frac{\mathrm{d}i}{\mathrm{d}t} = \frac{v_m}{L}\sin(\omega t)$$

Integrating with respect to time:

$$i(t) = \int \frac{v_m}{L}\sin(\omega t)\, \mathrm{d}t = -\frac{v_m}{\omega L}\cos(\omega t) + \text{constant}$$

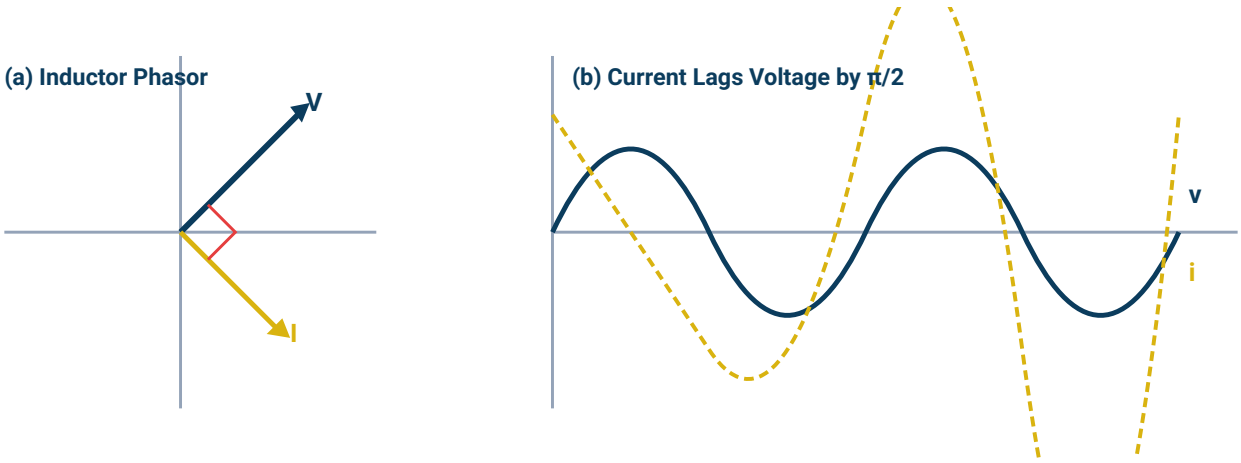
Because the applied source oscillates symmetrically about zero, no DC component exists, making the integration constant zero. Using $-\cos(\omega t) = \sin(\omega t - \pi/2)$:

$$i = i_m \sin\left(\omega t - \frac{\pi}{2}\right)$$

where the current amplitude is $i_m = \frac{v_m}{X_L}$, and X_L is defined as the **inductive reactance**:

$$X_L = \omega L = 2\pi\nu L$$

BOSONANCE



Tutorial Diagram 3: In an ideal inductor, alternating current lags voltage by $\pi/2$ radians (90°).

Inductive Reactance Properties:

- SI Unit: Ohm (Ω). Dimensional Formula: $[\mathrm{M\,L^2\,T^{-3}\,A^{-2}}]$.
- Proportionality: $X_L \propto \nu$ and $X_L \propto L$. For direct current ($\nu = 0$), $X_L = 0$; an inductor behaves as an ideal short circuit to DC in steady state.

Power Supplied to an Inductor: The instantaneous power is:

$$p_L = vi = [v_m \sin(\omega t)] \left[i_m \sin\left(\omega t - \frac{\pi}{2}\right)\right] = -v_m i_m \sin(\omega t)\cos(\omega t) = -\frac{v_m i_m}{2}\sin(2\omega t)$$

Taking the average over one complete cycle:

$$P_L = \langle p_L \rangle = -\frac{v_m i_m}{2}\langle \sin(2\omega t)\rangle = 0$$

Energy stored in the magnetic field during one quarter-cycle is returned entirely to the source in the subsequent quarter-cycle. *An ideal inductor consumes zero net electrical power over a cycle.*

2.5 AC Voltage Applied to a Pure Capacitor

When an alternating voltage $v = v_m \sin(\omega t)$ is connected across a capacitor of capacitance C, charge oscillates continuously between the plates. At any instant, $q(t) = Cv(t) = Cv_m \sin(\omega t)$. The resulting alternating current is:

$$i = \frac{\mathrm{d}q}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}[Cv_m \sin(\omega t)] = \omega C v_m \cos(\omega t)$$

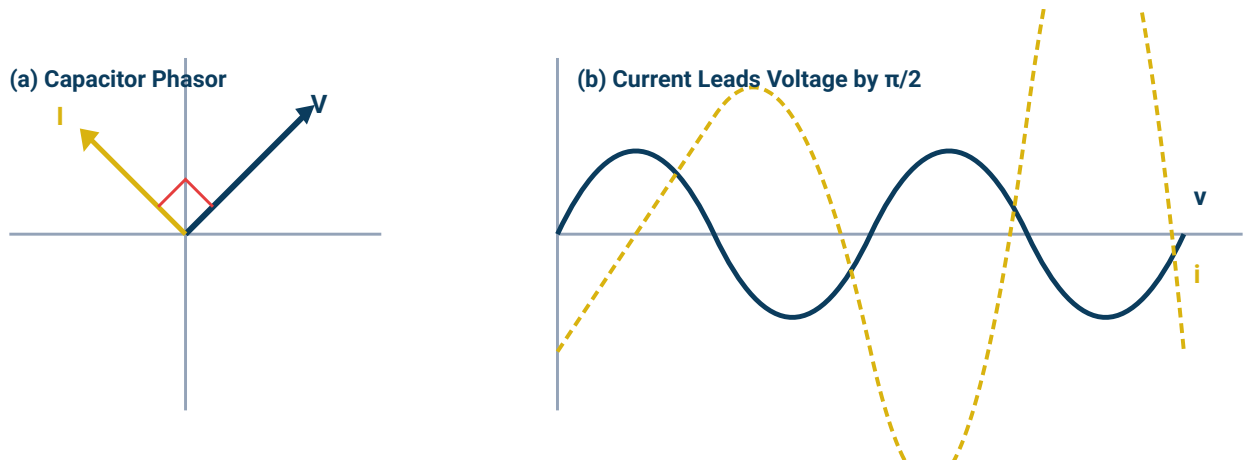
Using $\cos(\omega t) = \sin(\omega t + \pi/2)$:

$$i = i_m \sin\left(\omega t + \frac{\pi}{2}\right)$$

where $i_m = \frac{v_m}{X_C}$, and X_C is the **capacitive reactance**:

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi\nu C}$$

BOSONANCE



Tutorial Diagram 4: In an ideal capacitor, alternating current leads voltage by $\pi/2$ radians (90°).

Capacitive Reactance Properties:

- SI Unit: Ohm (Ω). Dimensional Formula: $[\mathrm{M\,L^2\,T^{-3}\,A^{-2}}]$.
- Proportionality: $X_C \propto \frac{1}{\nu}$ and $X_C \propto \frac{1}{C}$. For direct current ($\nu = 0$), $X_C \to \infty$; an ideal capacitor offers infinite opposition to DC and acts as an open circuit once charged.

Power Supplied to a Capacitor: The instantaneous power is:

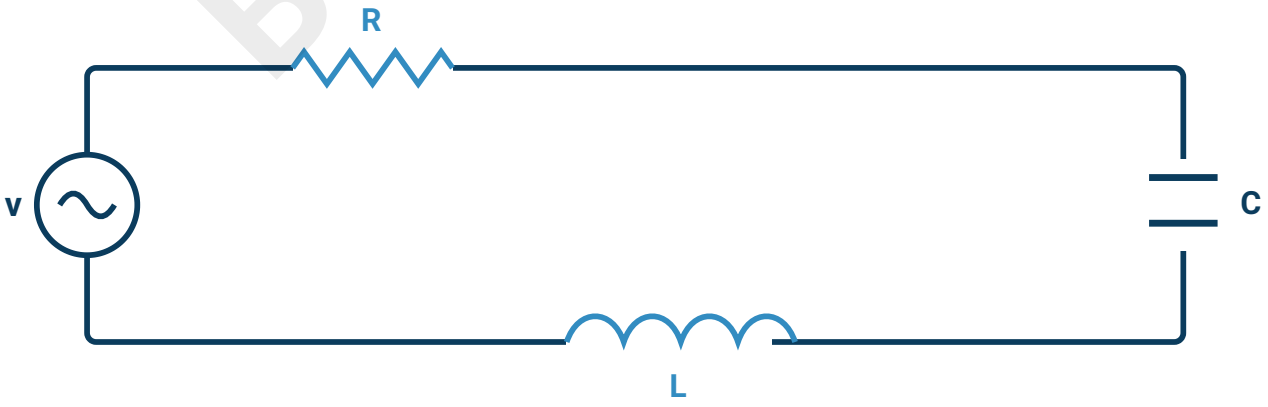
$$p_C = vi = [v_m \sin(\omega t)][i_m \cos(\omega t)] = \frac{v_m i_m}{2}\sin(2\omega t)$$

The average power over a complete cycle is zero:

$$P_C = \langle p_C \rangle = \frac{v_m i_m}{2}\langle \sin(2\omega t) \rangle = 0$$

3. AC Voltage Applied to a Series LCR Circuit

Consider a circuit where an inductor L, a capacitor C, and a resistor R are connected in series across an alternating source $v = v_m \sin(\omega t)$.



Tutorial Diagram 5: Series LCR circuit excited by a sinusoidal alternating emf.

At any instant t, let q be the charge on the capacitor and $i = \frac{\mathrm{d}q}{\mathrm{d}t}$ be the common series current. Kirchhoff's voltage loop equation gives:

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$$v = v_R + v_L + v_C \implies L\frac{\mathrm{d}i}{\mathrm{d}t} + iR + \frac{q}{C} = v_m \sin(\omega t)$$

3.1 Phasor Diagram Solution

Because all three circuit elements are wired in series, the identical alternating current flows through each of them:

$$i = i_m \sin(\omega t + \phi)$$

where ϕ is the phase difference between current and source voltage. Let $\mathbf{I}$ denote the current phasor. The individual element voltage phasors have amplitudes:

$$v_{Rm} = i_m R, \quad v_{Lm} = i_m X_L, \quad v_{Cm} = i_m X_C$$

- $\mathbf{V}_R$ is parallel to (in phase with) $\mathbf{I}$.
- $\mathbf{V}_L$ leads $\mathbf{I}$ by $\pi/2$ radians (90°).
- $\mathbf{V}_C$ lags $\mathbf{I}$ by $\pi/2$ radians (90°).

Since $\mathbf{V}_L$ and $\mathbf{V}_C$ lie along the same straight line in opposite directions, their resultant has magnitude $|v_{Cm} - v_{Lm}|$ or $|v_{Lm} - v_{Cm}|$. By vector addition of the perpendicular components $\mathbf{V}_R$ and $(\mathbf{V}_C + \mathbf{V}_L)$:

$$v_m^2 = v_{Rm}^2 + (v_{Cm} - v_{Lm})^2 = (i_m R)^2 + (i_m X_C - i_m X_L)^2$$

$$v_m = i_m \sqrt{R^2 + (X_C - X_L)^2}$$



Tutorial Diagram 6: Phasor geometry and impedance triangle for series LCR circuit.

3.2 Total Impedance and Phase Angle

By analogy with Ohm's law, we define the circuit's **total impedance** Z:

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

The current amplitude and RMS current are given by:

$$i_m = \frac{v_m}{Z}, \quad I = \frac{V}{Z}$$

From the impedance right triangle, the phase relationship ϕ satisfies:

$$\tan\phi = \frac{X_L - X_C}{R} \quad \text{or} \quad \tan\phi = \frac{X_C - X_L}{R} \quad (\text{depending on sign convention})$$

Three Operating Regimes of a Series LCR Circuit:

1. **Inductive Case ($X_L > X_C$):** Current lags voltage by phase angle ϕ. The circuit behaves predominantly inductively.

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2. **Capacitive Case ($X_C > X_L$):** Current leads voltage by phase angle ϕ. The circuit behaves predominantly capacitively.
3. **Resistive / Resonant Case ($X_L = X_C$):** $\phi = 0$; current and source voltage are precisely in phase.

4. Electrical Resonance & Sharpness of Tuning

In a series LCR circuit driven by a source of variable angular frequency ω, the reactance difference $(X_L - X_C)$ varies continuously. At a unique natural frequency ω_0, inductive reactance and capacitive reactance cancel each other completely:

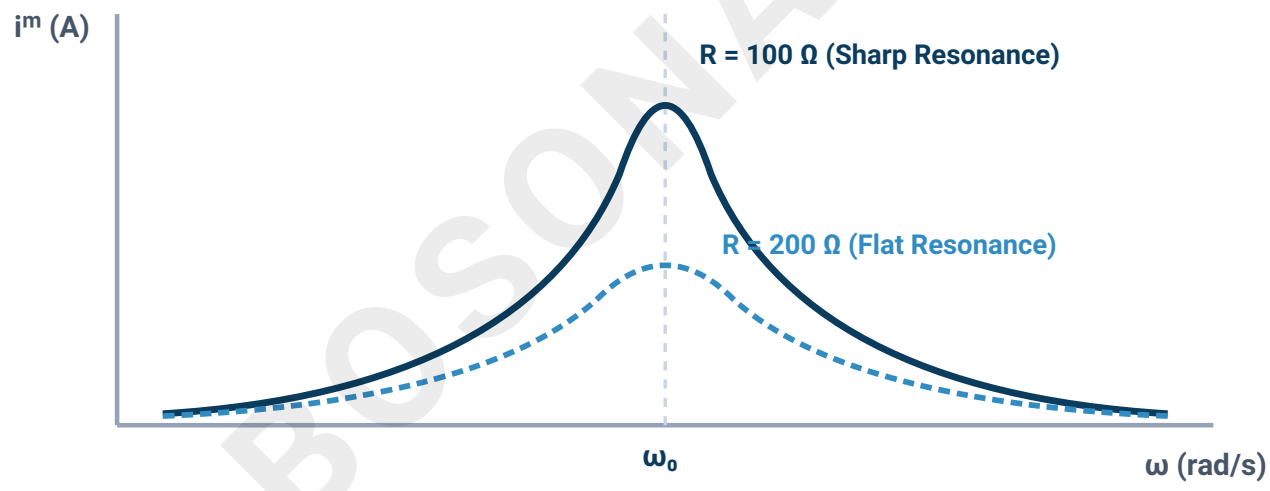
$$X_L = X_C \implies \omega_0 L = \frac{1}{\omega_0 C}$$

Solving for ω_0 and the resonant cyclic frequency ν_0:

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad \text{and} \quad \nu_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

At this resonant condition:

- The total impedance drops to its absolute theoretical minimum: $Z = \sqrt{R^2 + 0^2} = R$.
- The current amplitude achieves its peak maximum: $i_{m0} = \frac{v_m}{R}$.
- The voltages across L and C are equal in magnitude and 180° out of phase: $V_L + V_C = 0$. The entire source voltage drops across resistor R.



Tutorial Diagram 7: Current amplitude i_m versus driving angular frequency ω for two values of resistance R.

Essential NEET UG Conceptual Condition for Resonance:
Resonance can occur *only* in circuits possessing **both an inductor and a capacitor**. With only an RL or RC network, cancellation of reactive voltages is impossible; thus, genuine resonance cannot take place without both reactive elements.

4.1 Quality Factor (Q-Factor)

The sharpness of the resonance curve dictates the tuning selectivity of a radio or communications receiver. The dimensionless **Quality factor (Q)** is defined as:

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

BOSONANCE

A high Q-factor indicates a sharp resonance peak, allowing a receiver to select a desired station frequency while cleanly rejecting adjacent station broadcasts.

5. Power in AC Circuits & The Power Factor

For a series LCR circuit with source voltage $v = v_m \sin(\omega t)$ and current $i = i_m \sin(\omega t + \phi)$, the instantaneous power transferred is:

$$p = vi = [v_m \sin(\omega t)][i_m \sin(\omega t + \phi)] = \frac{v_m i_m}{2}[\cos\phi - \cos(2\omega t + \phi)]$$

Averaging over one complete cycle:

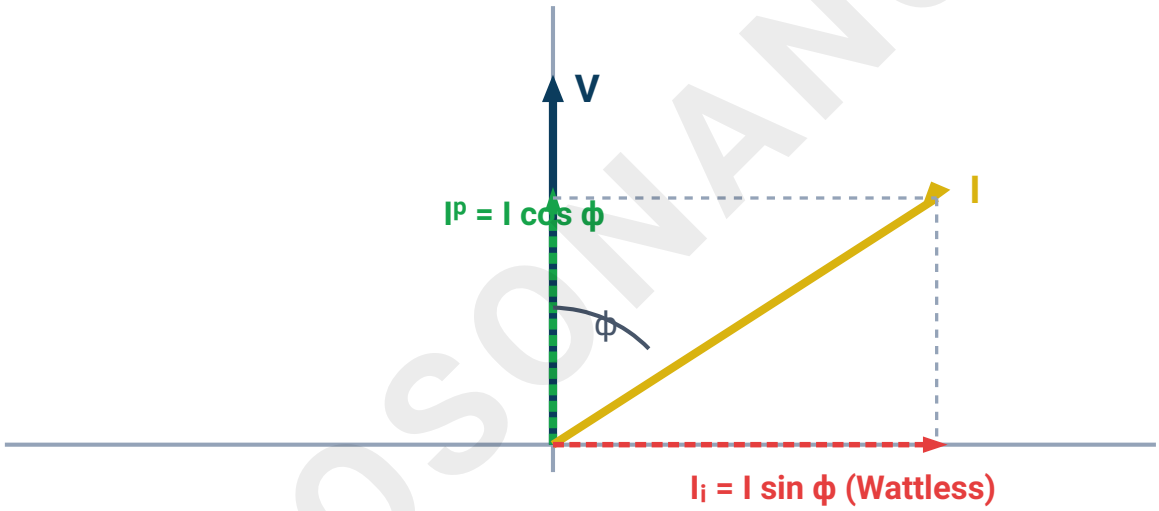
$$P = \frac{v_m i_m}{2}\cos\phi = \left(\frac{v_m}{\sqrt{2}}\right)\left(\frac{i_m}{\sqrt{2}}\right)\cos\phi = VI\cos\phi$$

Since $V = IZ$ and $\cos\phi = \frac{R}{Z}$, the average active power consumed is also:

$$P = I^2 Z\cos\phi = I^2 R$$

The dimensionless term $\cos\phi$ is called the **power factor** of the circuit:

$$\text{Power Factor } (\cos\phi) = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$$



Tutorial Diagram 8: Decomposition of AC current into active (work-producing) and reactive (wattless) components.

5.1 Wattless Current & Power Factor Correction

As depicted in Tutorial Diagram 8, the total current phasor $\mathbf{I}$ lagging voltage $\mathbf{V}$ by angle ϕ resolves into:

- **Active Component ($I_p = I\cos\phi$):** In phase with voltage; accounts for the actual energy dissipation ($P = VI_p$).
- **Reactive or Wattless Component ($I_q = I\sin\phi$):** Perpendicular (90°) to voltage. The power dissipated by this component is:

  $$P_q = V(I\sin\phi)\cos(90^\circ) = 0$$

  Because this component causes zero net power loss, it is termed **wattless current**.

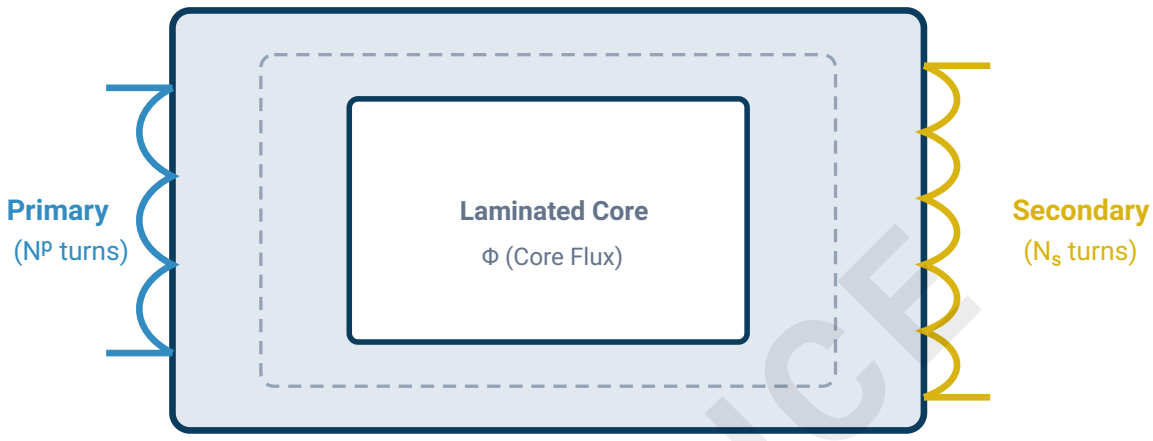
Why Power Companies Require Power Factor Correction:
To deliver a required power $P = VI\cos\phi$ to an industrial facility at standard distribution voltage V, a low power factor $\cos\phi$ forces the current $I = \frac{P}{V\cos\phi}$ to be significantly larger. This elevated current drastically magnifies transmission line Joule heating losses ($I^2 R_{\text{line}}$) and causes large voltage drops.

BOSONANCE

Consequently, industries connect **shunt capacitors in parallel** to supply a leading wattless current that neutralizes inductive lagging wattless currents, driving $\cos\phi \to 1$.

6. Transformers: Principle, Construction & Energetics

A **transformer** is a static electrical machine that steps alternating voltages up or down using the principle of **mutual induction**.



Tutorial Diagram 9: Schematic diagram of a transformer on a laminated soft-iron core.

6.1 Derivation of the Transformation Ratio

Let an alternating voltage v_p be connected across primary winding of N_p turns. By mutual induction, the time-varying current produces a flux Φ per turn in the iron core. By Faraday's Law, the induced back emf ε_p in the primary and induced secondary emf ε_s in the N_s turns of the secondary winding are:

$$\varepsilon_p = -N_p \frac{\mathrm{d}\Phi}{\mathrm{d}t}, \quad \varepsilon_s = -N_s \frac{\mathrm{d}\Phi}{\mathrm{d}t}$$

For an ideal transformer, winding resistance is negligible, meaning $v_p = \varepsilon_p$. With an open or high-impedance secondary load, $v_s = \varepsilon_s$. Taking their ratio:

$$\frac{v_s}{v_p} = \frac{N_s}{N_p}$$

For an ideal transformer with zero power loss (100% efficiency), input instantaneous power equals output instantaneous power:

$$p_{\text{in}} = p_{\text{out}} \implies i_p v_p = i_s v_s \implies \frac{i_p}{i_s} = \frac{v_s}{v_p} = \frac{N_s}{N_p}$$

Expressing this fundamental relation in terms of RMS terminal voltages and currents:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

Classification of Transformers:

- **Step-Up Transformer:** $N_s > N_p \implies V_s > V_p$ and $I_s < I_p$. Increases voltage while reducing current proportionally.
- **Step-Down Transformer:** $N_s < N_p \implies V_s < V_p$ and $I_s > I_p$. Lowers voltage while increasing current proportionally.

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- *Conservation of Energy:* A step-up transformer does not create energy; power gained in voltage is exactly balanced by the reduction in current.

6.2 Energy Losses in Real Transformers

Commercial transformers achieve exceptional efficiencies exceeding 95%, but experience four unavoidable physical losses:

Loss Mechanism	Physical Cause	Engineering Mitigation
Flux Leakage	Not all magnetic flux established by the primary links the secondary due to air gaps or leakage paths.	Winding primary and secondary coils coaxially one directly over the other on a closed core.
Resistance of Windings	Joule heating (I^2R) in copper conductor coils carrying high currents.	Using thick, high-conductivity copper wire for the high-current, low-voltage winding.
Eddy Currents	Alternating core flux induces closed circulating loop currents in the bulk metallic core, releasing heat.	Constructing the core from thin sheets of iron insulated with varnish (laminated core).
Hysteresis Loss	Continuous reversal of ferromagnetic domain magnetization during every AC cycle dissipates energy.	Using specialized soft magnetic materials with narrow B-H hysteresis loops (e.g., soft iron or silicon steel).

7. Summary Table of AC Physical Quantities

Physical Quantity	Symbol	Formula	SI Unit	Dimensional Formula
RMS Voltage	V or V_{rms}	$V = v_m/\sqrt{2}$	Volt (V)	$[\mathrm{M\,L^2\,T^{-3}\,A^{-1}}]$
RMS Current	I or I_{rms}	$I = i_m/\sqrt{2}$	Ampere (A)	$[\mathrm{A}]$
Inductive Reactance	X_L	$X_L = \omega L = 2\pi\nu L$	Ohm (Ω)	$[\mathrm{M\,L^2\,T^{-3}\,A^{-2}}]$
Capacitive Reactance	X_C	$X_C = \frac{1}{\omega C} = \frac{1}{2\pi\nu C}$	Ohm (Ω)	$[\mathrm{M\,L^2\,T^{-3}\,A^{-2}}]$
Series Impedance	Z	$Z = \sqrt{R^2 + (X_L - X_C)^2}$	Ohm (Ω)	$[\mathrm{M\,L^2\,T^{-3}\,A^{-2}}]$
Resonant Angular Frequency	ω_0	$\omega_0 = 1/\sqrt{LC}$	$\mathrm{rad/s}$	$[\mathrm{T^{-1}}]$
Quality Factor	Q	$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$	Dimensionless	$[\mathrm{M^0\,L^0\,T^0\,A^0}]$
Power Factor	$\cos\phi$	$\cos\phi = R/Z$	Dimensionless	$[\mathrm{M^0\,L^0\,T^0\,A^0}]$

8. Comprehensive Worked Examples

Worked Example 1: Purely Resistive Load

Problem: An electric incandescent lamp is rated at $100\ \mathrm{W}$ for a $220\ \mathrm{V}$, $50\ \mathrm{Hz}$ AC supply. Calculate: (a) the filament resistance of the lamp; (b) the peak voltage of the supply source; (c) the RMS current passing through the lamp.

Solution:

(a) Given rated power $P = 100\,\mathrm{W}$ and line voltage $V = 220\,\mathrm{V}$:

$$R = \frac{V^2}{P} = \frac{(220\,\mathrm{V})^2}{100\,\mathrm{W}} = \frac{48400}{100} = 484\,\Omega$$

(b) The peak voltage v_m is related to the RMS voltage by:

$$v_m = \sqrt{2} \times V \approx 1.414 \times 220\,\mathrm{V} \approx 311.08\,\mathrm{V} \approx 311\,\mathrm{V}$$

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(c) The RMS current through the lamp is:

$$I = \frac{P}{V} = \frac{100\,\text{W}}{220\,\text{V}} \approx 0.4545\,\text{A} \approx 0.455\,\text{A}$$

Worked Example 2: Pure Inductor Calculations

Problem: A pure inductor with inductance $L = 25.0\,\text{mH}$ is connected across a $220\,\text{V}$, $50\,\text{Hz}$ alternating line supply. Determine the inductive reactance and the RMS current in the circuit.

Solution:

Inductive reactance X_L is given by:

$$X_L = 2\pi\nu L = 2 \times 3.1416 \times 50\,\text{s}^{-1} \times (25.0 \times 10^{-3}\,\text{H}) = 314.16 \times 0.025\,\Omega = 7.854\,\Omega$$

The RMS current flowing in the pure inductor is:

$$I = \frac{V}{X_L} = \frac{220\,\text{V}}{7.854\,\Omega} \approx 28.01\,\text{A} \approx 28.0\,\text{A}$$

Worked Example 3: Pure Capacitor and Frequency Variation

Problem: A $15.0\,\mu\text{F}$ capacitor is connected to a $220\,\text{V}$, $50\,\text{Hz}$ AC source. (a) Find the capacitive reactance, the RMS current, and the peak current. (b) What happens to capacitive reactance and current if the frequency is doubled?

Solution:

(a) Capacitive reactance X_C:

$$X_C = \frac{1}{2\pi\nu C} = \frac{1}{2 \times 3.1416 \times 50 \times 15.0 \times 10^{-6}} = \frac{1}{4.7124 \times 10^{-3}} \approx 212.2\,\Omega$$

The RMS current is:

$$I = \frac{V}{X_C} = \frac{220\,\text{V}}{212.2\,\Omega} \approx 1.037\,\text{A} \approx 1.04\,\text{A}$$

The peak current i_m is:

$$i_m = \sqrt{2} \times I \approx 1.414 \times 1.037\,\text{A} \approx 1.47\,\text{A}$$

(b) Since $X_C = \frac{1}{2\pi\nu C}$, doubling frequency $\nu \to 2\nu$ halves the capacitive reactance ($X_C' = X_C/2 \approx 106.1\,\Omega$). Consequently, current doubles ($I' = 2I \approx 2.08\,\text{A}$).

Worked Example 4: Series RC Voltage Paradox Resolution

Problem: A resistor $R = 200\,\Omega$ and a capacitor $C = 15.0\,\mu\text{F}$ are connected in series to a $220\,\text{V}$, $50\,\text{Hz}$ source. (a) Compute the current in the circuit. (b) Compute the individual voltages across the resistor (V_R) and capacitor (V_C). (c) The algebraic sum of V_R and V_C is $151\,\text{V} + 160.3\,\text{V} = 311.3\,\text{V}$, which exceeds the $220\,\text{V}$ source. Explain this paradox.

Solution:

(a) Using $X_C \approx 212.3\,\Omega$, the total impedance Z is:

$$Z = \sqrt{R^2 + X_C^2} = \sqrt{(200)^2 + (212.3)^2} = \sqrt{40000 + 45071} = \sqrt{85071} \approx 291.67\,\Omega$$

$$I = \frac{V}{Z} = \frac{220\,\text{V}}{291.67\,\Omega} \approx 0.7543\,\text{A} \approx 0.755\,\text{A}$$

(b) The RMS potential differences across each element are:

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$$V_R = IR = 0.7543 \times 200 \approx 150.9\,\text{V} \approx 151\,\text{V}$$
$$V_C = IX_C = 0.7543 \times 212.3 \approx 160.1\,\text{V} \approx 160\,\text{V}$$

(c) **Paradox Resolution:** AC voltages cannot be summed algebraically like scalar numbers because they are not in phase. Voltage across the capacitor lags current by 90°, while voltage across the resistor is in phase with current. Hence, V_R and V_C are mutually perpendicular (90° phase difference). Vector (phasor) addition yields:

$$V_{\text{total}} = \sqrt{V_R^2 + V_C^2} = \sqrt{(150.9)^2 + (160.1)^2} = \sqrt{22771 + 25632} = \sqrt{48403} \approx 220\,\text{V}$$

The vector resultant strictly equals the source voltage.

Worked Example 5: Series LCR Circuit and Resonance

Problem: A sinusoidal voltage with peak value $v_m = 283\,\text{V}$ and frequency $\nu = 50\,\text{Hz}$ is applied to a series LCR circuit with $R = 3\,\Omega$, $L = 25.48\,\text{mH}$, and $C = 796\,\mu\text{F}$. (a) Calculate X_L, X_C, and total impedance Z. (b) Determine the phase angle ϕ and state whether current leads or lags voltage. (c) Determine the power dissipated in the circuit and the power factor. (d) Find the resonant frequency ν_0 and power dissipated at resonance.

Solution:

(a) Reactance calculations:

$$X_L = 2\pi(50)(25.48 \times 10^{-3}) = 314.16 \times 0.02548 \approx 8.0\,\Omega$$
$$X_C = \frac{1}{2\pi(50)(796 \times 10^{-6})} = \frac{1}{314.16 \times 7.96 \times 10^{-4}} \approx 4.0\,\Omega$$
$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{3^2 + (8-4)^2} = \sqrt{9+16} = \sqrt{25} = 5\,\Omega$$

(b) Phase angle:

$$\tan\phi = \frac{X_L - X_C}{R} = \frac{8-4}{3} = \frac{4}{3} \implies \phi = \arctan\left(\frac{4}{3}\right) \approx 53.1^\circ$$

Because $X_L > X_C$, the circuit is predominantly inductive, and **current lags the applied voltage by** 53.1°.

(c) Current and power:

$$i_m = \frac{v_m}{Z} = \frac{283}{5} = 56.6\,\text{A} \implies I = \frac{i_m}{\sqrt{2}} = \frac{56.6}{1.414} \approx 40\,\text{A}$$
$$\text{Power Factor } = \cos\phi = \frac{R}{Z} = \frac{3}{5} = 0.6$$
$$P = I^2 R = (40\,\text{A})^2 \times 3\,\Omega = 1600 \times 3 = 4800\,\text{W} = 4.8\,\text{kW}$$

(d) Resonant frequency and resonant dissipation:

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{25.48 \times 10^{-3} \times 796 \times 10^{-6}}} = \frac{1}{\sqrt{2.028 \times 10^{-5}}} \approx 222.1\,\text{rad/s}$$
$$\nu_0 = \frac{\omega_0}{2\pi} = \frac{222.1}{6.283} \approx 35.4\,\text{Hz}$$

At resonance, $Z = R = 3\,\Omega$. The resonant RMS current is:

$$I_{\text{res}} = \frac{V}{R} = \frac{283/\sqrt{2}}{3} = \frac{200\,\text{V}}{3\,\Omega} \approx 66.67\,\text{A}$$
$$P_{\text{res}} = I_{\text{res}}^2 R = (66.67\,\text{A})^2 \times 3\,\Omega \approx 13333\,\text{W} \approx 13.33\,\text{kW}$$

BOSONANCE

9. NEET UG Practice Questions with Worked Solutions

Practice Question 1

Single-Correct MCQ

A $100\,\Omega$ electric heating element is connected to an alternating source defined by $v = 200\sqrt{2}\sin(100\pi t)\ \text{V}$. The time required for the current to swing from its peak positive value to its peak negative value, and the net power consumed over a complete cycle are respectively:

(A) $0.01\ \text{s}$, $400\ \text{W}$

(B) $0.02\ \text{s}$, $200\ \text{W}$

(C) $0.01\ \text{s}$, $800\ \text{W}$

(D) $0.005\ \text{s}$, $400\ \text{W}$

Correct Option: (A)

Step-by-step Solution:

1. Angular frequency $\omega = 100\pi\ \text{rad/s}$. Therefore, cyclic frequency $\nu = \frac{\omega}{2\pi} = \frac{100\pi}{2\pi} = 50\ \text{Hz}$.
2. Time period of one full cycle $T = \frac{1}{\nu} = \frac{1}{50} = 0.02\ \text{s}$.
3. Current swings from positive maximum to negative maximum across half a period:

$$\Delta t = \frac{T}{2} = \frac{0.02}{2} = 0.01\ \text{s}$$

4. RMS voltage $V = \frac{v_m}{\sqrt{2}} = \frac{200\sqrt{2}}{\sqrt{2}} = 200\ \text{V}$.
5. Average active power consumed:

$$P = \frac{V^2}{R} = \frac{(200)^2}{100} = \frac{40000}{100} = 400\ \text{W}$$

Practice Question 2

Single-Correct MCQ

A pure inductor of self-inductance $L = 44\ \text{mH}$ is connected across a $220\ \text{V}$, $50\ \text{Hz}$ mains supply. The RMS current through the coil and the net electrical energy absorbed over one full cycle are:

(A) $15.9\ \text{A}$, $3500\ \text{J}$

(B) $15.9\ \text{A}$, zero

(C) $31.8\ \text{A}$, zero

(D) $7.95\ \text{A}$, $1750\ \text{J}$

Correct Option: (B)

Step-by-step Solution:

1. Reactance $X_L = 2\pi\nu L = 2 \times \frac{22}{7} \times 50 \times 44 \times 10^{-3} = \frac{2200 \times 44 \times 10^{-3}}{7} \approx 13.83\ \Omega$.
2. RMS current $I = \frac{V}{X_L} = \frac{220}{13.83} \approx 15.91\ \text{A} \approx 15.9\ \text{A}$.
3. For a purely inductive circuit, phase difference $\phi = 90^\circ$. Power factor $\cos\phi = \cos(90^\circ) = 0$.
4. Average power $P = VI\cos(90^\circ) = 0$. Net energy absorbed over a complete cycle is identically zero.

BOSONANCE

Practice Question 3

Single-Correct MCQ

A $60\,\mu\text{F}$ capacitor is connected to a $110\,\text{V}, 60\,\text{Hz}$ AC source. What is the RMS current flowing in the circuit?

(A) $1.24\,\text{A}$

(B) $2.49\,\text{A}$

(C) $3.73\,\text{A}$

(D) $0.62\,\text{A}$

Correct Option: (B)

Step-by-step Solution:

1. Capacitive reactance:

$$X_C = \frac{1}{2\pi\nu C} = \frac{1}{2 \times 3.1416 \times 60 \times 60 \times 10^{-6}} = \frac{1}{0.02262} \approx 44.21\,\Omega$$

2. RMS current:

$$I = \frac{V}{X_C} = \frac{110\,\text{V}}{44.21\,\Omega} \approx 2.488\,\text{A} \approx 2.49\,\text{A}$$

Practice Question 4

Single-Correct MCQ

A charged $30\,\mu\text{F}$ capacitor is connected across an ideal $27\,\text{mH}$ inductor. The natural angular frequency of free LC electromagnetic oscillations in the circuit is:

(A) $1.11 \times 10^3\,\text{rad/s}$

(B) $1.11 \times 10^2\,\text{rad/s}$

(C) $9.0 \times 10^2\,\text{rad/s}$

(D) $3.33 \times 10^3\,\text{rad/s}$

Correct Option: (A)

Step-by-step Solution:

1. Natural angular frequency of an LC oscillator:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

2. Substituting values:

$$LC = (27 \times 10^{-3}\,\text{H}) \times (30 \times 10^{-6}\,\text{F}) = 810 \times 10^{-9} = 81 \times 10^{-8}\,\text{s}^2$$

$$\omega_0 = \frac{1}{\sqrt{81 \times 10^{-8}}} = \frac{1}{9 \times 10^{-4}} = \frac{10000}{9} \approx 1111.1\,\text{rad/s} \approx 1.11 \times 10^3\,\text{rad/s}$$

BOSONANCE

Practice Question 5

Single-Correct MCQ

A series LCR circuit with $R = 20\,\Omega$, $L = 1.5\,\text{H}$, and $C = 35\,\mu\text{F}$ is driven by a variable frequency $200\,\text{V}$ AC source. When the source frequency matches the natural frequency of the circuit, what is the average power transferred over one complete cycle?

(A) $1000\,\text{W}$

(B) $2000\,\text{W}$

(C) $4000\,\text{W}$

(D) Zero

Correct Option: (B)

Step-by-step Solution:

1. When supply frequency equals the natural frequency, the circuit is at **resonance**.
2. At resonance, $X_L = X_C$, so impedance $Z = R = 20\,\Omega$.
3. Current at resonance: $I = \frac{V}{R} = \frac{200\,\text{V}}{20\,\Omega} = 10\,\text{A}$.
4. Average power transferred at resonance:

$$P = I^2 R = (10\,\text{A})^2 \times 20\,\Omega = 100 \times 20 = 2000\,\text{W}$$

Practice Question 6

Single-Correct MCQ

A series LCR circuit has $L = 5.0\,\text{H}$, $C = 80\,\mu\text{F}$, and $R = 40\,\Omega$ connected to a variable-frequency $230\,\text{V}$ source. The resonant angular frequency ω_0 and the current amplitude i_m at resonance are:

(A) $50\,\text{rad/s}, 8.13\,\text{A}$

(B) $25\,\text{rad/s}, 5.75\,\text{A}$

(C) $50\,\text{rad/s}, 5.75\,\text{A}$

(D) $100\,\text{rad/s}, 8.13\,\text{A}$

Correct Option: (A)

Step-by-step Solution:

1. Resonant angular frequency:

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{5.0 \times 80 \times 10^{-6}}} = \frac{1}{\sqrt{400 \times 10^{-6}}} = \frac{1}{20 \times 10^{-3}} = 50\,\text{rad/s}$$

2. At resonance, total impedance is strictly $Z = R = 40\,\Omega$.
3. Source voltage amplitude $v_m = \sqrt{2} \times 230\,\text{V} \approx 1.414 \times 230 \approx 325.22\,\text{V}$.
4. Current amplitude at resonance:

$$i_m = \frac{v_m}{R} = \frac{325.22\,\text{V}}{40\,\Omega} \approx 8.13\,\text{A}$$

(Note: The RMS current would be $I = \frac{230}{40} = 5.75\,\text{A}$, while current amplitude is $i_m = 8.13\,\text{A}$).

BOSONANCE

Practice Question 7

Assertion-Reason

Assertion (A): When an iron rod is inserted into the interior core of an inductor connected in series with a glowing light bulb across an AC source, the brightness of the bulb decreases.
Reason (R): Inserting an iron core decreases the self-inductance of the coil, thereby increasing inductive reactance.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) Both (A) and (R) are false.

Correct Option: (C)

Step-by-step Solution:
- Inserting a ferromagnetic iron rod concentrates the magnetic flux, dramatically *increasing* the coil's self-inductance L (not decreasing it). Hence, Reason (R) is false.
- As L increases, inductive reactance $X_L = \omega L$ increases. The total circuit impedance $Z = \sqrt{R_{\text{bulb}}^2 + X_L^2}$ increases, reducing circuit current $I = V/Z$.
- With reduced current, power dissipated in the lamp I^2R decreases, causing the brightness to drop. Thus, Assertion (A) is true.

BOSONANCE

Practice Question 8

Statement-Based Question

Consider the following statements regarding an ideal step-up transformer:
Statement I: A step-up transformer increases voltage while decreasing current, thus conserving electrical power.
Statement II: Laminating the soft-iron core with insulating lacquer prevents flux leakage between the primary and secondary coils.
Choose the correct evaluation:

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true, but Statement II is false.

(D) Statement I is false, but Statement II is true.

Correct Option: (C)

Step-by-step Solution:
- *Statement I is correct:* By $V_s I_s = V_p I_p$, stepping up voltage proportionately lowers current, adhering strictly to energy conservation.
- *Statement II is incorrect:* Core lamination breaks circulating electrical paths to suppress **eddy current heating**, not flux leakage. Flux leakage is minimized by coaxial winding geometry.

Practice Question 9

Single-Correct MCQ

In an AC circuit, the instantaneous voltage and current are given by $v = 100 \sin\left(100t + \frac{\pi}{3}\right)\ \text{V}$ and $i = 5 \sin(100t)\ \text{A}$. The power factor of the circuit and the average power consumed are:

(A) $0.5,\ 125\ \text{W}$

(B) $0.866,\ 216.5\ \text{W}$

(C) $0.5,\ 250\ \text{W}$

(D) $1.0,\ 500\ \text{W}$

Correct Option: (A)

Step-by-step Solution:
1. Phase difference $\phi = \left(100t + \frac{\pi}{3}\right) - (100t) = \frac{\pi}{3} = 60^\circ$.
2. Power factor $= \cos\phi = \cos(60^\circ) = 0.5$.
3. Peak values: $v_m = 100\,\text{V}$, $i_m = 5\,\text{A}$.
4. Average power consumed:

$$P = \frac{v_m i_m}{2}\cos\phi = \frac{100 \times 5}{2} \times 0.5 = 250 \times 0.5 = 125\,\text{W}$$

BOSONANCE

Practice Question 10

Single-Correct MCQ

An ideal transformer has 100 turns in its primary coil and 200 turns in its secondary coil. If the primary winding is supplied with $220\ \text{V}$ at a current of $10\ \text{A}$, the output voltage and secondary current are:

(A) $440\ \text{V}$, $5.0\ \text{A}$

(B) $110\ \text{V}$, $20.0\ \text{A}$

(C) $440\ \text{V}$, $20.0\ \text{A}$

(D) $110\ \text{V}$, $5.0\ \text{A}$

Correct Option: (A)

Step-by-step Solution:

1. Turns ratio: $\frac{N_s}{N_p} = \frac{200}{100} = 2$.
2. Secondary voltage:

$$V_s = \left(\frac{N_s}{N_p}\right) V_p = 2 \times 220\ \text{V} = 440\ \text{V}$$

3. Secondary current:

$$I_s = \left(\frac{N_p}{N_s}\right) I_p = \frac{1}{2} \times 10\ \text{A} = 5.0\ \text{A}$$

10. Common Pitfalls & High-Yield Revision Checklist

Trap Alerts for NEET UG Physics:

- **Peak vs. RMS Confusions:** Household socket voltage ($220\ \text{V}$ or $240\ \text{V}$) is *always* the RMS value unless explicitly specified as amplitude or peak. Peak voltage is $\sqrt{2} \times 220\ \text{V} \approx 311\ \text{V}$.
- **Algebraic vs. Vector Addition:** In an RC, RL, or LCR circuit, never sum individual voltmeter readings algebraically ($V \neq V_R + V_L + V_C$). You must use phasor geometry: $V = \sqrt{V_R^2 + (V_L - V_C)^2}$.
- **Average Power vs. Instantaneous Power:** The average power over a complete cycle in a pure inductor or pure capacitor is identically zero, even though instantaneous power fluctuates continuously between positive and negative values.
- **Frequency Dependence of Reactance:** For DC ($\nu = 0$), $X_L = 0$ (short circuit) and $X_C \to \infty$ (open circuit). In high-frequency limits ($\nu \to \infty$), an inductor blocks current ($X_L \to \infty$), while a capacitor passes current easily ($X_C \to 0$).
- **Resonance Requirement:** A circuit cannot exhibit resonance unless *both* L and C are present to cancel each other's reactive voltage.

Quick Formula Revision Sheet

- $V_{\text{rms}} = v_m/\sqrt{2} \approx 0.707\, v_m \quad \Longleftrightarrow \quad v_m = \sqrt{2} V_{\text{rms}} \approx 1.414\, V_{\text{rms}}$
- $X_L = \omega L = 2\pi\nu L \quad \text{and} \quad X_C = \frac{1}{\omega C} = \frac{1}{2\pi\nu C}$
- $Z = \sqrt{R^2 + (X_L - X_C)^2}, \quad \tan\phi = \frac{X_L - X_C}{R}, \quad \cos\phi = \frac{R}{Z}$
- Resonance: $\omega_0 = \frac{1}{\sqrt{LC}}, \quad \nu_0 = \frac{1}{2\pi\sqrt{LC}}, \quad Z_{\min} = R, \quad I_{\max} = \frac{V}{R}$

BOSONANCE

- Quality Factor: $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R}\sqrt{\frac{L}{C}}$
- Power Dissipation: $P = VI\cos\phi = I^2 R$
- Transformer Equations: $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$

11. Final Chapter Recap

Alternating current forms the backbone of modern energy distribution. Resistors dissipate energy continuously with current and voltage exactly in phase ($\phi = 0$). Inductors oppose current variations via induced back emf, causing current to lag voltage by $\pi/2$ (90°), while capacitors accumulate charge such that current leads voltage by $\pi/2$. In series LCR circuits, total opposition is represented by impedance Z. At electrical resonance ($X_L = X_C$), the opposing reactive voltages cancel completely, impedance collapses to pure resistance R, and current reaches its maximum possible amplitude. Power dissipation in any AC circuit occurs solely in the resistive components, with average active power governed by the power factor $\cos\phi = R/Z$. Finally, mutual induction in transformers allows voltages to be stepped up or down efficiently, enabling long-distance power transmission that powers our world.

BOSONANCE